

Quantum Operations, Quantum Noise

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1. Quantum Operations

Exercise 8.1: (Unitary evolution as a quantum operation) [cite_start]Pure states evolve under unitary transforms as $|\psi\rangle \rightarrow U|\psi\rangle$. [cite: 22] [cite_start]Show that, equivalently, we may write $\rho \rightarrow \varepsilon(\rho) \equiv U\rho U^\dagger$, for $\rho = |\psi\rangle\langle\psi|$. [cite: 23]

Solution:

1. [cite_start]If pure states evolve as $|\psi\rangle \rightarrow U|\psi\rangle$, then their duals evolve like so $\langle\psi| \rightarrow \langle\psi|U^\dagger$. [cite: 24, 25]

2. [cite_start]Now consider how ρ would evolve: [cite: 25] [cite_start]

$$\rho = |\psi\rangle\langle\psi| \rightarrow U|\psi\rangle\langle\psi|U^\dagger = U\rho U^\dagger$$

. [cite: 26]

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Exercise 8.2: (Measurement as a quantum operation)

[cite_start]Recall from Section 2.2.3 (on page 84) that a quantum measurement with outcomes labeled by m is described by a set of measurement operators M_m such that $\sum_m M_m^\dagger M_m = I$. [cite: 27] Let the state of the system immediately before the measurement be ρ . [cite_start]Show that for $\varepsilon_{m(\rho)} \equiv M_m \rho M_m^\dagger$, the state of the system immediately after the measurement is [cite: 28]

$$\frac{\varepsilon_m(\rho)}{\text{tr}(\varepsilon(\rho))}$$

[cite_start]Also show that the probability of obtaining this measurement result is $p(m) = \text{tr}(\varepsilon_m(\rho))$. [cite: 28]

[cite_start]Solution: [cite: 29] First let's show the post measurement state of the density operator.

1. [cite_start]Note that $\varepsilon_{m(\rho)} \equiv M_m \rho M_m^\dagger = M_m |\psi\rangle\langle\psi| M_m^\dagger$. [cite: 29]

2. [cite_start]Post measurement state is $\frac{M_m |\psi\rangle\langle\psi|}{\sqrt{\langle\psi| M_m^\dagger M_m |\psi\rangle}}$. [cite: 29, 30]

3. Post measurement state of the dual is $\frac{\langle\psi| M_m^\dagger}{\sqrt{\langle\psi| M_m^\dagger M_m |\psi\rangle}}$. (Note: Denominator inside square root typically involves $M^\dagger M$, corrected from source text logic where applicable)[cite_start]. [cite: 30, 31]

4. Thus, the post measurement state of ρ is $\frac{M_m |\psi\rangle\langle\psi| M_m^\dagger}{\langle\psi| M_m^\dagger M_m |\psi\rangle}$. [cite: 31, 32, 33]

5. [cite_start]Note that the numerator is $\varepsilon_{m(\rho)}$ and that the denominator is $\text{tr}(\varepsilon_{m(\rho)})$. [cite: 33]

Now, let's see the probability.

1. [cite_start]Note that the probability of measuring m after measurement is given by $p(m) = \langle\psi| M_m^\dagger M_m |\psi\rangle$. [cite: 34, 35]

2. [cite_start]Note that this is $\text{tr}(\varepsilon(\rho))$. [cite: 35]

■

¹Assuming the measurement operator commutes with its adjoint. [cite_start]Does it?